Where the rotation frequency of the sync block comes from

proposed block (trained) · trained · no parameters

fixed (final)

ω = 2π / P
P: one learned period
(init 24 sampled frames)
the same for every clip

↓

rotation of state pairs
cos, sin(k_j·ω·t), k = 1, 2, 1, 2

an LTI complex-diagonal SSM;
no heart-rate estimate needed
(robust on short clips)

VAL MAE 3.963

hr

ω = periodogram peak of the
clip's own frame features
(period grid 6–48 frames,
label-free, no gradient)

↓

rotation of state pairs
cos, sin(k_j·ω·t), k = 1, 2, 1, 2

adapts to each heart rate;
needs ≥ ~1 beat in the clip

VAL MAE 3.982

tempered

ω = (2π / P) · (ω_hr / ω_ref)^γ
γ = sigmoid(learned), init 0.5
ω_ref = 2π / 22.5 (median)
clips < 24 frames: ratio = 1

↓

rotation of state pairs
cos, sin(k_j·ω·t), k = 1, 2, 1, 2

γ = 0 → fixed, γ = 1 → hr
(the learned γ stayed ≈ 0.5)

VAL MAE 3.985

learned (Mamba-3 style)

angle_t = Σ_{s ≤ t} π·tanh(W x_s)
one angle per state pair,
from the local input
(data-dependent, per step)

↓

rotation of state pairs
by the accumulated angle

replaces k_j·ω·t

VAL MAE 4.046

none

no rotation
(real decay only,
Mamba-2-like SSD)

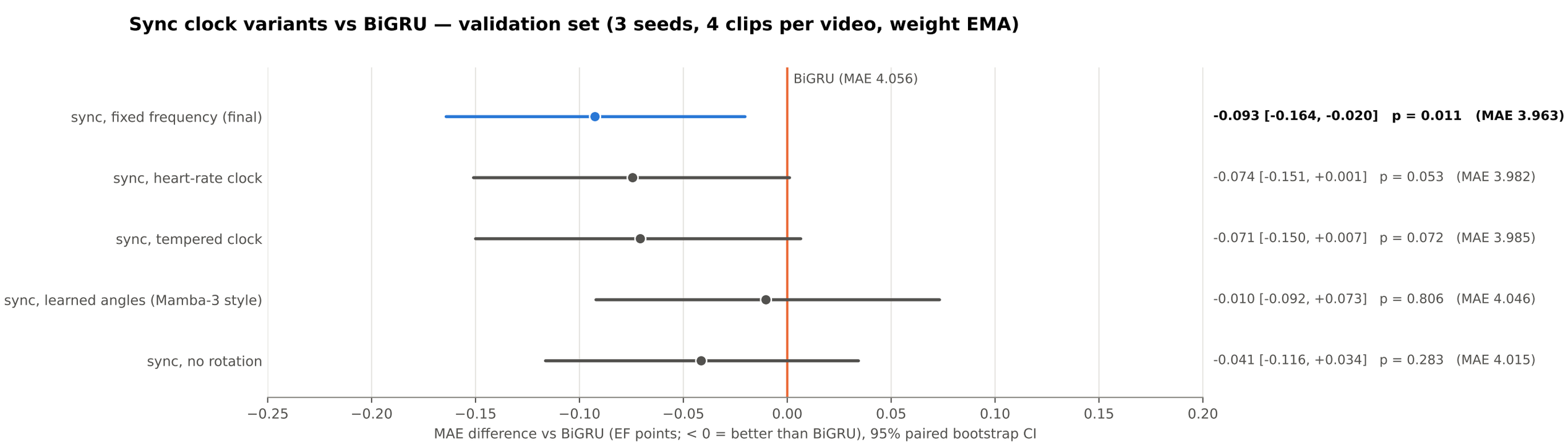
↓

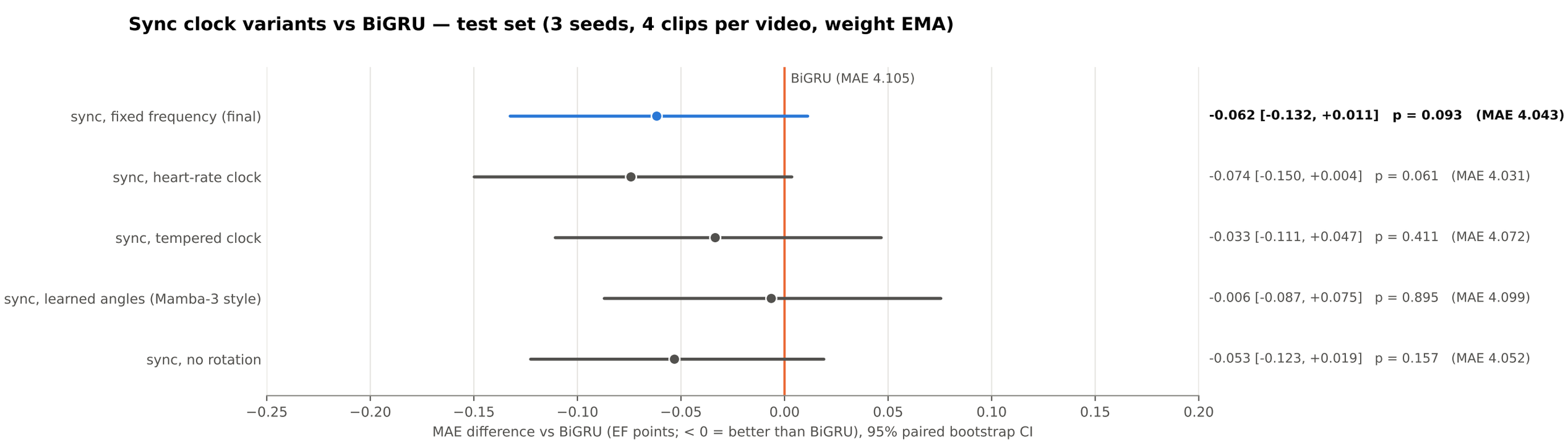
identity
(no rotation)

same parameters as
the other variants

VAL MAE 4.015

VAL MAE: 3 seeds, 4 clips per video, frozen PanEcho 3-frame-stack features, weight EMA. Everything else in the block is identical across the five variants.


Sync clock variants vs BiGRU — validation set (3 seeds, 4 clips per video, weight EMA)
BiGRU (MAE 4.056)
sync, fixed frequency (final)
sync, heart-rate clock
sync, tempered clock
sync, learned angles (Mamba-3 style)
sync, no rotation
-0.093 [-0.164, -0.020] p = 0.011 (MAE 3.963)
-0.074 [-0.151, +0.001] p = 0.053 (MAE 3.982)
-0.071 [-0.150, +0.007] p = 0.072 (MAE 3.985)
-0.010 [-0.092, +0.073] p = 0.806 (MAE 4.046)
-0.041 [-0.116, +0.034] p = 0.283 (MAE 4.015)
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
MAE difference vs BiGRU (EF points; < 0 = better than BiGRU), 95% paired bootstrap CI


Sync clock variants vs BiGRU — test set (3 seeds, 4 clips per video, weight EMA)
BiGRU (MAE 4.105)
sync, fixed frequency (final)
sync, heart-rate clock
sync, tempered clock
sync, learned angles (Mamba-3 style)
sync, no rotation
-0.062 [-0.132, +0.011] p = 0.093 (MAE 4.043)
-0.074 [-0.150, +0.004] p = 0.061 (MAE 4.031)
-0.033 [-0.111, +0.047] p = 0.411 (MAE 4.072)
-0.006 [-0.087, +0.075] p = 0.895 (MAE 4.099)
-0.053 [-0.123, +0.019] p = 0.157 (MAE 4.052)
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
MAE difference vs BiGRU (EF points; < 0 = better than BiGRU), 95% paired bootstrap CI